A Qubit No-Go Bound for Independent Commuting Finite-Order Controls

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Two finite-order quantum controls may each be compatible with a qubit while their joint algebra is not. We study a cyclic control Q and an independent involution R with target algebra $C_n \times C_2$. For even $n \geq 4$, no faithful projective qubit realization exists: in $PU(2) \simeq SO(3)$, a nontrivial involution commuting with an order-n rotation collapses to $Q^{n/2}$. A qutrit realizes the full algebra exactly, so the minimum projective dimension is three; $n = 4$ is the smallest obstructed case and has an eight-state orbit. More importantly, the obstruction yields a tight operational qubit tradeoff. For an exact quarter-turn Q and half-turn R, if their commutator has Bloch rotation angle γ_C, then every qubit obeys $F(U_R\psi, U_Q^2\psi) \geq \cos(\gamma_C/2)$, equivalently a tight binary-contrast cap $\Delta \leq \sqrt{1-\cos(\gamma_C/2)}$. We derive a fully angular-error version and extend it to general CPTP gate deviations through diamond distance. An exact qutrit construction reaches the algebraic maximum: with one seed state, the same binary measurement distinguishes the two sequences R and Q^2 with probability contrast $\Delta = 1$ while Q and R commute exactly. The result is a calibrated low-setting falsification test for a qubit control model, not a device-independent dimension witness.

I. INTRODUCTION

The dimension of a quantum system can be certified from operational data in several settings, including prepare-and-measure protocols, sequential measurements, and repeated dynamics [1–4]. A related line of work uses algebraic relations and approximate representations to infer structural properties of quantum states or subspaces [5, 6]. These developments motivate a simple question: can a small set of finite-order control relations themselves impose a minimum Hilbert-space dimension?

Here we study the Abelian control algebra

$$G_n = C_n \times C_2 = \langle Q, R \mid Q^n = R^2 = \mathbb{I},\ QR = RQ \rangle, \quad (1)$$

with the additional requirement that the two factors act independently on quantum rays. Operationally, Q labels a cyclic orientation coordinate and R an independent binary reversal. The mathematically elementary exact problem has a sharp physical consequence: for even $n \geq 4$, a qubit cannot support a faithful projective action of Eq. (1); dimension three can.

The exact representation-theory statement alone is not our main purpose. Minimal faithful (projective) dimensions of finite groups belong to a mature mathematical literature [7]. Instead, we use the obstruction to derive a tight quantitative qubit consistency inequality and reduce its experimental test to two control sequences followed by the same binary measurement. The qutrit construction reaches unit binary contrast, giving the largest possible separation from the exactly commuting qubit case. This places the result near, but distinct from, repeated-operation dimension tests [4], robust algebraic self-testing [6], group-theoretic unitary-property testing [8], and recent information-processing tasks on state families generated by group actions [10].

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II. EXACT FINITE-ORDER OBSTRUCTION

We first work with reversible controls and pure-state rays. A reversible CPTP channel on a fixed finite-dimensional system whose inverse is also CPTP is a unitary conjugation. In particular, if a CPTP map obeys $\mathcal{E}^k = \mathrm{id}$ for finite k, then $\mathcal{E}^{-1} = \mathcal{E}^{k-1}$ is CPTP and the map is unitary. Thus exact finite-order channel relations reduce to projective unitary representations.

a. Theorem 1. For even $n \geq 4$, the group $C_n \times C_2$ has no faithful projective qubit representation, whereas it has a faithful projective qutrit representation. Hence its minimum projective quantum dimension is three. The case $n = 4$ is the smallest such even-n obstruction and has an eight-state regular orbit.

b. Proof. Projective qubit unitaries form $PU(2) \simeq SO(3)$. Let the image of Q have order $n \geq 4$. It is a nontrivial rotation about some axis. Every rotation in $SO(3)$ that commutes with such an element must preserve the same axis; hence a commuting element of order two is the half-turn about that axis. For even n, the half-turn generated by the same cyclic subgroup is precisely $Q^{n/2}$ (up to the choice of generator). Therefore the putatively independent C_2 factor is identified with an element of C_n, so the projective representation is not faithful.

A faithful qutrit realization is explicit. Let

$$Q_n = \mathrm{diag}(1, e^{2\pi i/n}, 1), \qquad R = \mathrm{diag}(1, 1, -1). \quad (2)$$

Then $Q_n^n = R^2 = \mathbb{I}$ and $Q_n R = R Q_n$. If $Q_n^a R^b$ is proportional to the identity, its first diagonal entry fixes the proportionality factor to one, after which $e^{2\pi i a/n} = 1$ and $(-1)^b = 1$. Thus only $(a, b) = (0, 0)$ lies in the projective kernel. □

A convenient seed with all three amplitudes nonzero is

$$|\psi_*\rangle = \frac{|0\rangle}{\sqrt{2}} + \frac{|1\rangle}{2} + \frac{|2\rangle}{2}. \quad (3)$$

Its orbit is

$$|\psi_{a,b}\rangle = \frac{|0\rangle}{\sqrt{2}} + \frac{e^{2\pi i a/n}|1\rangle}{2} + \frac{(-1)^b|2\rangle}{2}, \tag{4}$$

with $2n$ distinct rays. For $n = 4$, this is the minimal eight-state member of the family. The weights in Eq. (3) are chosen so that the independent involution and the cyclic half-turn produce orthogonal output states, maximizing the binary separation derived below.

For odd n, $C_n \times C_2 \simeq C_{2n}$ is cyclic and admits a qubit projective realization. For $n = 2$, $C_2 \times C_2$ is the Klein four group, which also embeds in $SO(3)$. Thus $n = 4$ is the first case in this family where the independent binary factor forces a dimension jump.

III. QUBIT TRADEOFF UNDER APPROXIMATE CLOSURE

We next derive a quantitative version for $n = 4$. Let the target qubit unitaries U_Q, U_R correspond on the Bloch sphere to rotations by angles θ_Q and θ_R about axes separated by α. Let the commutator

$$C = U_Q U_R U_Q^{-1} U_R^{-1} \tag{5}$$

have principal Bloch rotation angle γ_C.

A direct $SU(2)$ calculation gives the scalar part of a lift of the commutator,

$$s_C = 1 - 2\sin^2\frac{\theta_Q}{2}\sin^2\frac{\theta_R}{2}\sin^2\alpha, \tag{6}$$

while its principal Bloch rotation angle obeys $\cos(\gamma_C/2) = |s_C|$. For an exact quarter-turn and half-turn, $\theta_Q = \pi/2$ and $\theta_R = \pi$, Eq. (6) gives $s_C = \cos^2\alpha \geq 0$, and hence

$$\cos\frac{\gamma_C}{2} = \cos^2\alpha. \tag{7}$$

The relative unitary between U_R and U_Q^2 is then a rotation whose minimum pure-state fidelity is $\cos^2\alpha$. Hence, for every pure qubit state,

$$F(U_R\psi, U_Q^2\psi) \geq \cos\frac{\gamma_C}{2}. \tag{8}$$

The bound is tight: for any allowed axis angle one can choose the seed Bloch vector perpendicular to the axis of the relative unitary $U_R^\dagger U_Q^2$. Consequently, optimizing over a common binary measurement gives the tight qubit contrast bound

$$\Delta_{\text{bin}}^{(U)} \leq \sqrt{1 - \cos(\gamma_C/2)}. \tag{9}$$

If F_C denotes the average gate fidelity of the qubit commutator C with the identity, then $F_C = (2 + \cos\gamma_C)/3$ and

$$F(U_R\psi, U_Q^2\psi) \geq \sqrt{\frac{3F_C - 1}{2}}. \tag{10}$$

Thus exact commutation forces U_R and U_Q^2 to act identically on rays and forces the binary contrast to zero.

A useful fully approximate form follows if

$$|\theta_Q - \pi/2| \leq \delta_Q, \qquad |\theta_R - \pi| \leq \delta_R, \qquad \gamma_C \leq \delta_C, \tag{11}$$

with $0 \leq \delta_Q < \pi/2$, $0 \leq \delta_R < \pi$, and the sufficient branch condition

$$\delta_Q + \delta_C/2 < \pi/2. \tag{12}$$

The last condition excludes the negative-$SU(2)$-lift branch of a projectively small commutator: throughout the allowed window Eq. (6) has $s_C \geq -\sin\delta_Q$, whereas $\gamma_C \leq \delta_C$ requires $|s_C| \geq \cos(\delta_C/2)$. Thus $s_C = \cos(\gamma_C/2)$ in the relevant branch, and Eq. (6) implies

$$\sin^2\alpha \leq B, \quad B = \frac{1 - \cos(\delta_C/2)}{2\sin^2(\pi/4 - \delta_Q/2)\cos^2(\delta_R/2)}. \tag{13}$$

Let $\bar{B} = \min\{1, B\}$. Using the scalar part of the relative $SU(2)$ rotation between U_R and U_Q^2 gives the state-independent bound

$$F(U_R\psi, U_Q^2\psi) \geq F_{\min}, \tag{14}$$

$$F_{\min} = \left[\cos\frac{\delta_R}{2}\cos\delta_Q\sqrt{1-\bar{B}} - \sin\frac{\delta_R}{2}\sin\delta_Q\right]_+^2. \tag{15}$$

For $\delta_Q = \delta_R = \delta_C = 5°$, Eq. (15) gives $F_{\min} \simeq 0.98193$.

The same geometric calculation extends to the even family. For exact orders and even $n \geq 4$,

$$F(U_R\psi, U_Q^{n/2}\psi) \geq \left[1 - \frac{1 - \cos(\gamma_C/2)}{2\sin^2(\pi/n)}\right]_+. \tag{16}$$

At fixed commutator error, the right-hand side is strongest at $n = 4$. Thus the eight-state member is not only the smallest obstructed orbit but also the most robust member of this family against a fixed commutator-angle error.

IV. GENERAL CPTP DEVIATIONS AND A BINARY OBSERVABLE

Let the experimentally implemented channels $\widetilde{Q}$ and $\widetilde{R}$ obey

$$D_\diamond(\widetilde{Q}, \mathcal{U}_Q) \leq \eta_Q, \qquad D_\diamond(\widetilde{R}, \mathcal{U}_R) \leq \eta_R, \tag{17}$$

where $D_\diamond(\mathcal{E}, \mathcal{F}) = \frac{1}{2}\|\mathcal{E} - \mathcal{F}\|_\diamond$. By contractivity and composition,

$$D_\diamond(\widetilde{Q}^2, \mathcal{U}_Q^2) \leq 2\eta_Q. \tag{18}$$

For any qubit state ρ, convexity of trace distance and Eq. (15) give

$$D_{\text{tr}}\left[\widetilde{R}(\rho), \widetilde{Q}^2(\rho)\right] \leq \sqrt{1 - F_{\min}} + \eta_R + 2\eta_Q. \tag{19}$$

Consequently, for any common binary POVM effect $0 \le M \le \mathbb{I}$, the probability contrast

$$\Delta_{\rm bin} = \left|\mathrm{Tr}[M\widetilde{R}(\rho)] - \mathrm{Tr}[M\widetilde{Q}^2(\rho)]\right| \tag{20}$$

satisfies

$$\boxed{\Delta_{\rm bin} \le \sqrt{1-F_{\rm min}} + \eta_R + 2\eta_Q.} \tag{21}$$

Equation (21) is a calibrated qubit consistency inequality: it is not device independent, but it is independent of the unknown qubit input state and of the common binary measurement.

V. MAXIMAL QUTRIT VIOLATION WITH TWO SEQUENCES

We now specialize Eq. (4) to $n = 4$:

$$Q = \mathrm{diag}(1, i, 1), \qquad R = \mathrm{diag}(1, 1, -1). \tag{22}$$

The controls commute exactly, have the required finite orders, and generate eight distinct rays from Eq. (3). The two relevant outputs are

$$R|\psi_*\rangle = \frac{|0\rangle}{\sqrt{2}} + \frac{|1\rangle}{2} - \frac{|2\rangle}{2}, \tag{23}$$

$$Q^2|\psi_*\rangle = \frac{|0\rangle}{\sqrt{2}} - \frac{|1\rangle}{2} + \frac{|2\rangle}{2}, \tag{24}$$

and they are exactly orthogonal:

$$\langle\psi_*|R^\dagger Q^2|\psi_*\rangle = 0. \tag{25}$$

Choose the fixed binary effect

$$M_* = R|\psi_*\rangle\langle\psi_*|R^\dagger. \tag{26}$$

Then

$$p_*(R) = 1, \qquad p_*(Q^2) = 0, \tag{27}$$

so the qutrit reaches the algebraic maximum

$$\boxed{\Delta_{\rm qutrit} = 1.} \tag{28}$$

Thus exact commutation gives a zero contrast for the qubit target algebra but allows unit contrast in dimension three.

For imperfect controls, Eqs. (15) and (21) provide the calibrated qubit cap. For example, with $\delta_Q = \delta_R = \delta_C = 5^\circ$ and no additional CPTP gate deviation, the qubit contrast is at most $\sqrt{1-F_{\rm min}} \simeq 0.1344$, while the ideal qutrit value is one. Supplemental Material [12] gives detailed derivations, a conservative conversion from reported average gate fidelity to a diamond-distance budget, and additional verification details.

For N independent shots on each branch, Hoeffding's inequality gives

$$\Pr\left[|\widehat{\Delta} - \Delta| \ge t\right] \le 2e^{-Nt^2}. \tag{29}$$

Hence a sufficient rejection rule at significance α is

$$\widehat{\Delta} > \tau_Q + \sqrt{\frac{\ln(2/\alpha)}{N}}, \tag{30}$$

where τ_Q is the calibrated qubit cap from Eq. (21).

VI. RELATION TO EXISTING CERTIFICATION SCHEMES

The present result occupies a narrower niche than general dimension witnessing. Prepare-and-measure witnesses certify dimension from families of preparations and measurements [1, 2]; sequential schemes can certify dimensions above two or three [3]; and a single repeated operation can be used in a delay/rank null test [4]. Our constraint is instead generated by a pair of finite-order controls whose individual actions are compatible with a qubit but whose *independent commuting joint action* is not.

Approximate algebraic relations have also been used to certify ground-space degeneracy and to construct robust self-tests [5, 6]. Those works establish broad and powerful representation-stability principles. Our contribution is the explicit low-dimensional $C_n \times C_2$ control obstruction, its minimal $n = 4$ member, the closed-form commutator-versus-separation tradeoff, and the reduction to a two-sequence binary contrast with a qutrit realization.

Recent work has combined group representation theory with optimal testing of unitary properties [8], studied conclusive exclusion for quantum-state families generated by finite and compact group actions [10], and developed channel-dimensionality certification under minimal assumptions [9]. The exclusion problem asks which group-orbit alternatives can be ruled out with zero error; our task instead asks whether the underlying joint control algebra itself can be realized by a qubit. Recent trapped-ion experiments have also demonstrated robust certification of non-projective measurements in two- and three-dimensional systems [11], illustrating that calibrated low-dimensional qudit certification is experimentally accessible while addressing a distinct property of the measurement device. These tasks are complementary to Eq. (21), which is a calibrated model-consistency test for the dimensionality required to realize a specified joint control algebra.

VII. DISCUSSION

The main message is that an apparently mild Abelian algebra can contain a dimension-sensitive joint constraint. A qubit supports quarter-turns and half-turns separately,

but if an order-four control and an independent order-two control are required to commute faithfully, the Bloch-sphere centralizer forces the binary factor to collapse into the cyclic subgroup. A third level is the smallest way to keep the two factors independent.

The eight-state orbit is therefore not introduced as an ontological subdivision of a qubit. It is the minimal regular orbit of the first obstructed member, $C_4 \times C_2$, of a finite-order control family. This distinction is essential: the result is a statement about realizable quantum control algebras and operational statistics, not about substructure below a qubit.

Several limitations remain. The robust inequality assumes independently calibrated bounds on the target rotation classes and on gate deviations. It is therefore not a fully device-independent dimension witness. A useful next step is to replace some calibration assumptions by direct operational relation tests and to derive confidence regions under correlated and non-Markovian noise. The present construction is deliberately minimal: two control sequences and one common binary readout suffice for the ideal unit-contrast separation. Experimental implementation on a controllable qutrit would test whether that large calibrated gap survives realistic control errors.

DATA AVAILABILITY

No experimental data were generated or analyzed in this work. All analytic results required for the conclusions are contained in the article and Supplemental Material. Verification scripts and numerical stress-test records used only as consistency checks are included in the accompanying reproducibility archive.

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[12] See Supplemental Material accompanying this preprint for the projective-centralizer argument, exact qutrit representation, $SU(2)$ derivations, CPTP-noise chaining, average-fidelity conversion, AI-assisted verification workflow, and finite-shot details.

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